

SOCIOL 723: Social Statistics II

Week 9, Regression Discontinuity Designs

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Today

Sharp Regression Discontinuity

Estimation

Covariates and Power

Validity

Local Randomization

Fuzzy RD

Application: Bombing in Vietnam

Extensions

Applications in Sociology

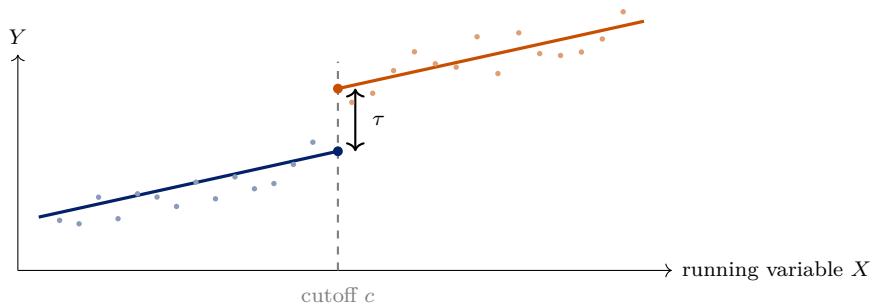
Practice

★ = essential, you must understand this

★ = for understanding, not examined

The Idea in One Picture ★

Many treatments are assigned by a rule: a test score threshold, an income cut-off, a vote share above 50%, an age of eligibility.



Just below and just above the cutoff, units are essentially identical, except that one group got the treatment. The jump in the outcome at c is the effect.

Where RD Sits

Goal: an effect estimate when treatment is assigned by a rule on a measured score, with no overlap anywhere.

design	comparable units	needs
matching, weighting (Week 6)	same X , both treatments observed	overlap, $0 < e(x) < 1$
IV (Week 7)	moved by Z , which moves only D	exclusion, monotonicity
RD (today)	just either side of a cutoff in X	continuity at the cutoff

Plan:

1. Sharp RD: why continuity identifies the effect at c , and only there.
2. Estimation: bias, variance, bandwidth, and valid intervals.
3. Covariates and power; validity checks; local randomization.
4. Fuzzy RD as local IV; Dell and Querubín (2018) on bombing in Vietnam.
5. Extensions: several cutoffs, away from the cutoff, boundaries, kinks, time.
6. Two sociological applications.

Sharp RD

Setup and Identification ★

A **running variable** (or forcing variable) X_i determines treatment deterministically:

$$D_i = \mathbb{1}\{X_i \geq c\}.$$

Write $m_d(x) = E[Y_i(d) \mid X_i = x]$ for the two potential-outcome curves.

Continuity assumption (Hahn, Todd, and van der Klaauw 2001)

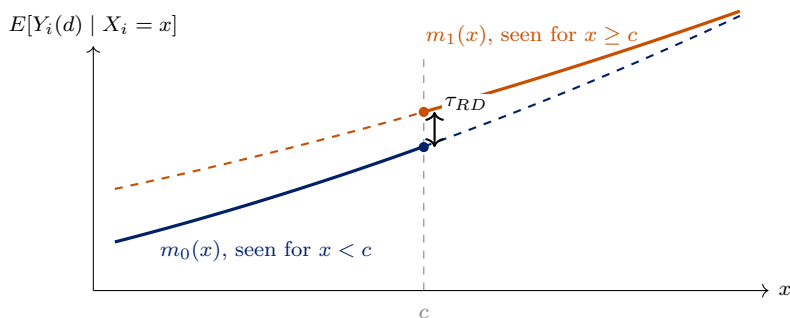
$m_1(x)$ and $m_0(x)$ are continuous in x at $x = c$.

Then

$$\tau_{RD} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x] = E[Y_i(1) - Y_i(0) \mid X_i = c].$$

- Not assumed: anything about the shape of $E[Y \mid X]$ away from the cutoff, or ignorability anywhere else.
- **Positivity fails at every x :** $\Pr(D = 1 \mid X = x)$ is 0 or 1. Identification comes from continuity, not overlap, which is why the estimand is local to c .

Two Curves, Each Seen on One Side



- The observed regression $E[Y_i | X_i = x]$ follows m_0 to the left of c and m_1 to the right; it jumps at c by $m_1(c) - m_0(c)$.
- The dashed halves are never observed. Away from c the gap between the curves, $\tau(x)$, is not identified without further assumptions.

Identification, Step by Step

Above the cutoff every unit is treated, so $Y_i = Y_i(1)$ there:

$$\lim_{x \downarrow c} E[Y_i | X_i = x] = \lim_{x \downarrow c} E[Y_i(1) | X_i = x] \quad D_i = 1 \text{ for } x \geq c; \text{ consistency}$$

$$= \lim_{x \downarrow c} m_1(x) = m_1(c) \quad \text{continuity of } m_1 \text{ at } c$$

$$\lim_{x \uparrow c} E[Y_i | X_i = x] = \lim_{x \uparrow c} E[Y_i(0) | X_i = x] \quad D_i = 0 \text{ for } x < c; \text{ consistency}$$

$$= m_0(c) \quad \text{continuity of } m_0 \text{ at } c$$

Subtract:

$$\lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x] = m_1(c) - m_0(c) = E[Y_i(1) - Y_i(0) | X_i = c].$$

- Continuity of m_1 is used only from the right and of m_0 only from the left: we need the *unseen* side of each curve to meet the seen side at c .

What Continuity Rules Out

Continuity fails when something other than D changes at c .

- **A second treatment at the same cutoff.** An age threshold that grants a pension and also lowers a tax; a district boundary where schools, police, and taxes all change. The jump is the effect of the bundle.
- **Sorting.** If people who would do well anyway can place themselves just above c , the units just above differ from those just below in $Y_i(0)$ itself, and m_0 jumps.
- **Selection of the cutoff on outcomes.** An agency that sets c where it expects a change, after seeing the data, builds the jump in.

Continuity is a statement about $Y_i(0)$ and $Y_i(1)$, so it cannot be tested directly. It has **two testable implications**: the density of X and the distribution of pre-determined covariates should not jump at c . Both come from the next frame.

Why Continuity Should Hold (Lee 2008)

Let W_i be a unit's type (everything fixed before X is realized) and let X_i have density $f(x | w)$ given $W_i = w$. By Bayes' rule

$$f(w | x) = \frac{f(x | w) f(w)}{f(x)} \quad \text{density of types among units with } X_i = x$$

$$m_d(x) = \int E[Y_i(d) | W_i = w] f(w | x) dw \quad Y_i(d) \text{ is a function of type}$$

- If every type's density $f(x | w)$ is continuous at c , then $f(w | x)$ is continuous, so m_0 , m_1 , and the mean of any pre-determined covariate are continuous at c .
- This is **imprecise control**: units may influence X (study harder, campaign more), but a random component they cannot steer remains. Near c , treatment is as good as randomly assigned.
- It fails when some type can place itself exactly: then $f(x | w)$ jumps at c for that type, and so does $f(x)$, the density of X .

The Estimand Is Local, and That Matters ★

τ_{RD} is the average effect **for units at the cutoff**. Nothing more.

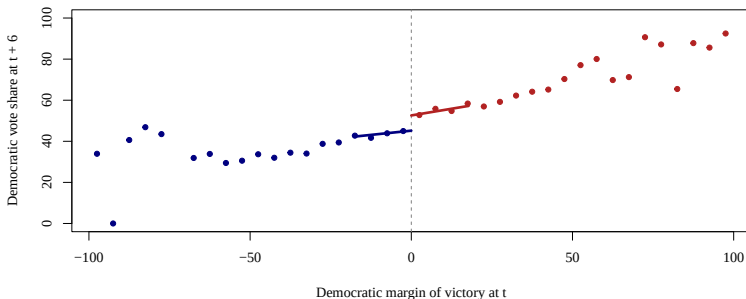
- A scholarship RD tells you about students at the eligibility margin, not about the strongest or weakest applicants.
- Extrapolating away from c requires a functional-form assumption of exactly the kind RD was designed to avoid.
- Compare with the LATE from Week 7: both are estimands you did not choose, determined by the design rather than by the question. State them plainly.
- In exchange, RD is often the most credible observational design there is. The identifying variation is visible in a scatterplot.

Lee (2008)'s reformulation is helpful: if agents cannot perfectly control X_i , then treatment near the cutoff is *as good as randomly assigned*, and the design is a local experiment.

Running Example: the U.S. Senate

Does winning a Senate seat help the party keep it six years later? (Lee 2008 for the House; Cattaneo, Frandsen, and Titiunik 2015 for the Senate)

- X_i : Democratic margin of victory at t , in points ($c = 0$); Y_i : Democratic vote share in the same seat at $t + 6$.
- 1,297 elections, 1914–2010 (595 lost, 702 won).



Dots: means in 5-point bins. Lines: the local linear fits used below (figs/w9_senate.R).

Estimation

Estimation: Local Linear Regression ★

Fit separate linear regressions on each side of the cutoff, using only observations within a bandwidth h , weighted by a kernel:

$$\hat{\tau} = \hat{\mu}_+ - \hat{\mu}_-, \quad (\hat{\mu}_+, \hat{\beta}_+) = \arg \min \sum_{i: c \leq X_i < c+h} K\left(\frac{X_i - c}{h}\right) (Y_i - \mu - \beta(X_i - c))^2,$$

and symmetrically below.

- **Local linear, not global polynomial.** High-order global polynomials give bizarre weights to distant observations and are now strongly discouraged (Gelman and Imbens 2019). Use degree 1 or 2.
- **Triangular kernel** (weights falling linearly to zero at h) is the standard choice and is boundary-optimal.
- Always let the slope differ on the two sides; forcing a common slope imposes a restriction you have no reason to believe.

The Estimator, Written Out

The RD estimate is the treatment coefficient in a **kernel-weighted, fully interacted local linear regression** on the observations within the bandwidth:

$$\min_{\mu, \tau, \beta_-, \beta_+} \sum_{i: |X_i - c| \leq h} K\left(\frac{X_i - c}{h}\right) \left(Y_i - \mu - \tau D_i - \beta_-(X_i - c)(1 - D_i) - \beta_+(X_i - c)D_i\right)^2,$$

with $D_i = \mathbb{1}\{X_i \geq c\}$ and $\hat{\tau}$ the estimate.

- **Centre the running variable at c** , so that the intercept shift τ is evaluated at the cutoff rather than at $X = 0$.
- **Interact the slope with D_i** . Forcing a common slope imposes a restriction you have no reason to believe, and it lets data on one side contaminate the estimate on the other.
- **Triangular kernel** $K(u) = (1 - |u|)_+$ weights observations linearly down to zero at the boundary and is minimax optimal at a boundary point. A uniform kernel is simpler and usually similar.

The Estimator on the Senate Data

`rdrobust` defaults: local linear, triangular kernel, $h = 17.75$ points (`figs/w9_senate.R`).

weighted <code>lm(vote ~ D * margin)</code> , $ X_i \leq 17.75$	
intercept = $\hat{\mu}_-$ (lost by a hair)	45.17
$D = \hat{\tau}$	7.41
slope below, $\hat{\beta}_-$	0.164
slope above, $\hat{\beta}_+ = 0.164 + 0.097$	0.261
<code>rdrobust</code> conventional estimate	7.41
effective n : below, above	360, 323

- A party that barely wins a seat gets $\hat{\mu}_+ = 52.6\%$ of the vote six years later; one that barely loses gets 45.2%.
- 683 of 1,297 elections carry positive weight; the rest do not enter the estimate.

Why Not Compare Means Near the Cutoff?

Take the mean of Y within h on each side (a local constant fit). Each OLS line passes through its side's means, so with $\bar{x}_{\pm}$ the mean distance to c :

$$\begin{aligned}\bar{Y}_+ - \bar{Y}_- &= (\hat{\mu}_+ + \hat{\beta}_+ \bar{x}_+) - (\hat{\mu}_- + \hat{\beta}_- \bar{x}_-) && \text{the line through the means} \\ &= \hat{\tau}_{\text{linear}} + \underbrace{\hat{\beta}_+ \bar{x}_+ - \hat{\beta}_- \bar{x}_-}_{\text{slope} \times \text{distance}} && \text{regroup}\end{aligned}$$

Senate, $h = 17.75$, uniform kernel:

$$10.91 = 7.09 + 0.260 \times 7.96 + 0.226 \times 7.75 = 7.09 + 2.07 + 1.75.$$

- **The bias of comparing means is of order h :** the means sit at $\pm h/2$ on average, not at c , and any slope turns that distance into a gap. Here it inflates the effect by half.
- A boundary is the worst place for a local mean: data lie on one side only, so the slope cannot cancel. Local linear removes the first-order term.

The Bias of Local Linear Is of Order h^2

Above c , write $u_i = (X_i - c)/h \in [0, 1]$; take u uniform (a uniform kernel, flat density) and expand

$$m_1(c + hu) = m_1(c) + m'_1(c) hu + \frac{1}{2}m''_1(c) h^2 u^2 + \dots \quad \text{Taylor.}$$

The line fits the first two terms exactly. The bias is the intercept of the best line through u^2 :

$$\text{slope} = \frac{\text{Cov}(u, u^2)}{\text{Var}(u)} = \frac{\frac{1}{4} - \frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{12}} = 1 \quad Eu^k = 1/(k+1)$$

$$\text{intercept} = Eu^2 - 1 \cdot Eu = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6} \quad \text{line through the means}$$

$$E[\hat{\mu}_+] - m_1(c) \approx \frac{1}{2}m''_1(c) h^2 \cdot \left(-\frac{1}{6}\right) = -\frac{1}{12} m''_1(c) h^2 \quad \text{scale by } \frac{1}{2}m''_1 h^2$$

- **Bias** = Bh^2 , with B set by the curvature on each side: $B = -\frac{1}{12} (m''_1(c) - m''_0(c))$. Other kernels change only the constant $\frac{1}{12}$.

The Variance Is of Order $1/(nh)$

With $f(c)$ the density of X at c , about $m = nf(c)h$ observations fall in $[c, c + h]$. The fitted intercept is a prediction at the *edge* of the data:

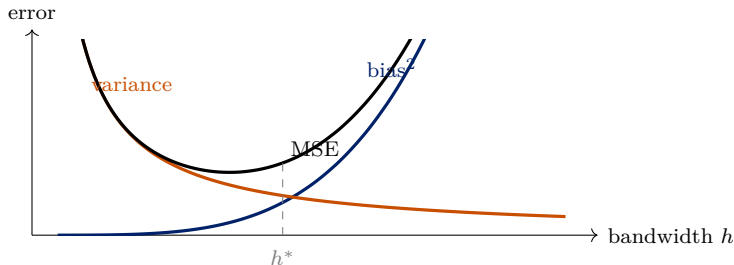
$$\text{Var}(\hat{\mu}_+) = \frac{\sigma^2}{m} \left(1 + \frac{(0 - \bar{u})^2}{\text{Var}(u)} \right) \quad \text{variance of an OLS prediction at } u = 0$$

$$= \frac{\sigma^2}{m} \left(1 + \frac{1/4}{1/12} \right) = \frac{4\sigma^2}{m} \quad u \text{ uniform on } [0, 1]$$

$$\text{Var}(\hat{\tau}) = \frac{4(\sigma_+^2 + \sigma_-^2)}{nf(c)h} \equiv \frac{V}{nh} \quad \text{two independent sides}$$

- **Extrapolating to the edge costs a factor of 4:** a mean at the centre of the window would have variance σ^2/m .
- The variance falls as the window widens, the bias rises: that is the bandwidth problem.

The Bandwidth Is the Whole Game ★



- **Wide h :** more data, lower variance, but the linear approximation is worse far from c , bias.
- **Narrow h :** less bias, but few observations, variance.
- Choose h by minimizing estimated MSE (Imbens–Kalyanaraman; the Calonico–Cattaneo–Titiunik version in `rdrobust`).
- The MSE-optimal bandwidth leaves a bias that invalidates the conventional interval. Use the CCT *robust bias-corrected* interval, which `rdrobust` reports by default.

The MSE-Optimal Bandwidth

Put the two previous frames together:

$$\text{MSE}(h) = B^2 h^4 + \frac{V}{nh} \quad \text{bias}^2 + \text{variance}$$

$$0 = 4B^2 h^3 - \frac{V}{nh^2} \quad \text{derivative in } h \text{ set to zero}$$

$$h^* = \left(\frac{V}{4B^2} \right)^{1/5} n^{-1/5} \quad \text{solve}$$

- **Wider** when the noise V is large or the density $f(c)$ is low; **narrower** when the curves bend sharply at c (B large).
- B and V are unknown: a pilot fit estimates m'' , σ^2 , and $f(c)$ and plugs them in (Imbens and Kalyanaraman 2012; Calonico, Cattaneo, and Titiunik 2014).
- h^* shrinks slowly: ten times the data narrows it by $10^{1/5} = 1.6$.
- Senate: $\hat{h}^* = 17.75$ points on both sides.

Why the Optimal Bandwidth Breaks the Confidence Interval ★

At h^* , the first-order condition fixes the ratio of bias to standard error:

$$4B^2h^{*5} = \frac{V}{n} \implies \underbrace{B^2h^{*4}}_{\text{bias}^2} = \frac{1}{4} \underbrace{\frac{V}{nh^*}}_{\text{variance}} \implies |\text{bias}| = \frac{1}{2} \text{se.} \quad \text{divide by } 4h^*$$

- The bias does not vanish relative to the standard error as n grows: both shrink like $n^{-2/5}$.
- $\hat{\tau} \pm 1.96 \hat{\text{se}}$ is centred half a standard error off: coverage $\Pr(|N(\frac{1}{2}, 1)| \leq 1.96) = 0.921$, lower when h is estimated. Covering 95% needs 2.18, not 1.96 (Armstrong and Kolesár 2020).
- The best h for a *point estimate* is not the best h for an *interval*.

Repairs: a smaller h (undersmoothing); subtract an estimate of the bias (robust bias correction); or bound the bias and widen by the bound (honest intervals).

Robust Bias Correction

Calonico, Cattaneo, and Titiunik (2014):

$$\hat{\tau}^{bc} = \hat{\tau}_h - \hat{B} h^2 \quad \hat{B}: \text{local quadratic, pilot bandwidth } b$$

$$\text{Var}(\hat{\tau}^{bc}) = \text{Var}(\hat{\tau}_h) + h^4 \text{Var}(\hat{B}) - 2h^2 \text{Cov}(\hat{\tau}_h, \hat{B}) \quad \hat{B} \text{ adds noise}$$

- The **robust** interval centres on $\hat{\tau}^{bc}$ and uses the second line; “bias-corrected” keeps the old standard error and undercovers.
- **Identity:** with $b = h$, $\hat{\tau}^{bc}$ and its robust s.e. are the local quadratic estimate at h and its s.e.

Senate, <code>rdrobust</code>	estimate	s.e.	95% interval
conventional, $h = 17.75$	7.41	1.46	[4.56, 10.27]
bias-corrected, $b = 28.03$	7.51	1.46	[4.65, 10.37]
robust	7.51	1.74	[4.09, 10.92]
robust with $b = h$	8.3212	2.0649	
local quadratic at h , conventional	8.3212	2.0649	

Does the Repair Work? A Simulation

The design of Calonico, Cattaneo, and Titiunik (2014): quintic curves fitted to Lee's House data, $\tau = 0.04$, 1,000 draws (figs/w9_sims.R).

coverage of a nominal 95% interval	$n = 500$	$n = 5,000$
conventional, MSE-optimal h	0.896	0.784
robust, MSE-optimal h	0.917	0.889
robust, CER-optimal h	0.930	0.935
bias / sd of the conventional estimate	0.31	0.80

- The conventional interval gets *worse* with more data.
- **CER-optimal bandwidth** (Calonico, Cattaneo, and Farrell 2020): a smaller h chosen to minimize the coverage error of the robust interval; $h_{CER} = h_{MSE} \times n^{-1/20}$ for local linear. Senate: $17.75 \times 0.699 = 12.41$.
- Practice: report the point estimate at h_{MSE} and the robust interval at h_{CER} (`bwselect = "cerrd"`).

Honest Intervals (Armstrong and Kolesár 2018, 2020)

Assume a bound on curvature, $|m_d''(x)| \leq M$ near c . For a given h the estimator's weights are known, so its **largest possible bias** $\bar{b}$ is computable. Then

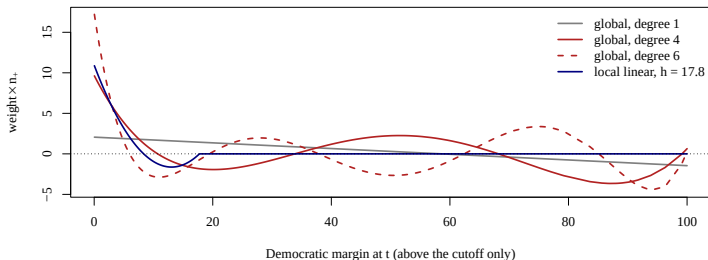
$$\hat{\tau} \pm \text{cv}_{0.95}(\bar{b}/\widehat{\text{se}}) \cdot \widehat{\text{se}}, \quad \text{cv}_{0.95}(t) = \text{the 95th percentile of } |N(t, 1)|.$$

- Coverage is at least 95% for every curve with $|m''| \leq M$: the interval is widened by exactly enough to absorb the worst bias.
- h is chosen to make the interval shortest; nothing is bias-corrected.
- The price is choosing M . A rule of thumb uses a global quartic fit; report how the interval changes with M .
- The same logic handles a **discrete running variable**, where continuity-based asymptotics do not apply (Kolesár and Rothe 2018).

Senate (RDHonest, $\hat{M} = 0.114$): $h = 9.84$, estimate 8.03, s.e. 1.85, $\bar{b} = 1.13$, $\text{cv} = 2.27$, interval $[3.82, 12.24]$.

Global Polynomials Weight Distant Races

Gelman and Imbens (2019). Every estimator of μ_+ is a weighted mean of the Y_i above the cutoff, weights = row 1 of $(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}$.



- A degree-6 fit gives a race won by 75 points 3.4 times the average weight; the estimate moves with the degree: $p = 1, \dots, 6$ gives 6.04, 4.93, 7.32, 9.41, 7.96, 6.63.
- The local linear weights are zero beyond h and largest at the cutoff.

Bandwidth Sensitivity on the Senate Data

Conventional estimate and robust 95% interval (b/h fixed at the default)

(figs/w9_senate.R):

	h	estimate	robust 95% interval	effective n (below, above)
	5	12.27	[7.93, 19.77]	128, 117
	10	7.98	[4.39, 13.11]	245, 206
12.41	(h_{CER})	7.63	[4.07, 11.29]	284, 248
17.75	(h_{MSE})	7.41	[4.09, 10.92]	360, 323
	30	7.28	[5.13, 10.71]	474, 457
	50	6.48	[4.87, 9.70]	565, 562

- Stable between $h = 10$ and 50; every interval excludes zero.
- At $h = 5$ the estimate rises, with 245 elections: noise, and the interval says so.

Covariates

Adding Covariates (Calonico, Cattaneo, Farrell, and Titiunik 2019)

Add pre-determined Z_i (lagged vote, say) to the local regression, additively, same kernel and bandwidth. By the omitted-variable formula (Week 1):

$$\tilde{\tau} = \hat{\tau}_Y - \hat{\tau}_Z' \hat{\gamma} \quad \text{exact, for weighted least squares}$$

$\hat{\tau}_Y$: RD estimate for Y ; $\hat{\tau}_Z$: RD estimates with each Z as the outcome; $\hat{\gamma}$: coefficients on Z in the regression that includes Z .

- If Z is continuous at c , $\tau_Z = 0$ and $\tilde{\tau}$ estimates the **same** τ_{RD} ; the gain is precision when Z predicts Y .
- If $\tilde{\tau}$ moves far from $\hat{\tau}_Y$, then $\hat{\tau}_Z$ is large: a covariate jumps at the cutoff. **That is a failed balance test, not an improvement.**
- Do not interact Z with D_i ; never use post-treatment Z .

Covariates on the Senate Data

Four pre-determined covariates, 1,205 complete elections, $h = 17.75$, triangular kernel (figs/w9_senate.R):

covariate Z	jump $\hat{\tau}_Z$	coefficient $\hat{\gamma}$
Democratic vote share, $t - 6$	4.047	0.112
Democratic vote share, $t - 12$	1.679	0.138
Democratic presidential vote share	-1.115	-0.019
open seat	-0.139	-1.119

$$\tilde{\tau} = 7.826 - (4.047 \cdot 0.112 + 1.679 \cdot 0.138 + (-1.115)(-0.019) + (-0.139)(-1.119)) = 6.966$$

- With `rdrobust`'s bandwidths: $7.82 \rightarrow 6.97$; robust interval 2% shorter.
- The shift comes mostly from the lagged vote, which jumps by 4 points ($p = 0.09$ in the balance test). Report both estimates.

Flexible Covariate Adjustment

Goal: more precision from many covariates without adding bias.

- Linear adjustment is one choice of a function $\eta(Z_i)$ subtracted from Y_i before the RD:

$$\tau_{RD} = \lim_{x \downarrow c} E[Y_i - \eta(Z_i) \mid X_i = x] - \lim_{x \uparrow c} E[Y_i - \eta(Z_i) \mid X_i = x],$$

true for *any* η when Z is continuous at c : the η terms cancel at the cutoff.

- So η can be estimated by machine learning, on held-out folds, to predict Y as well as possible; the estimand does not change (Noack, Olma, and Rothe 2026).
- The precision gain is largest when Z explains much of Y near the cutoff, such as a lagged outcome.

Week 13 returns to cross-fitting.

Why RD Needs Several Times the Data of an Experiment

Compare $\text{Var}(\hat{\tau})$ in RD with a coin-flip experiment, same n , same σ^2 , model

$$Y_i = \alpha + \tau D_i + \beta X_i + \epsilon_i:$$

$$\text{Var}(\hat{\tau}_{RD}) = \frac{\sigma^2}{n \text{Var}(D) (1 - \rho_{DX}^2)} \quad \text{FWL: only the part of } D \text{ not explained by } X \text{ counts}$$

$$\text{Var}(\hat{\tau}_{\text{exp}}) = \frac{\sigma^2}{n \text{Var}(D)} \quad D \text{ independent of } X$$

$$\text{ratio} = \frac{1}{1 - \rho_{DX}^2} \quad \rho_{DX} = \text{corr}(\mathbb{1}\{X \geq c\}, X)$$

X	ρ_{DX}^2	ratio	simulation (20,000 draws)
uniform, c at the middle	$3/4$	4	3.98
normal, c at the mean (Goldberger 1972)	$2/\pi$	2.75	2.73

Separate slopes: 4.07 (uniform), 2.73 (normal); local fits use fewer observations still (figs/w9_sims.R).

Covariates, Power, and What to Report

- **Covariates are for precision only.** They must be *pre-determined*, they should enter additively (not interacted with D), and the estimate should barely move when you add them. If it moves a lot, the design is in trouble, not improved (Calonico et al. 2019).
- **Effective sample size:** report the number of observations *inside the bandwidth* on each side. An RD with $n = 100,000$ often has 400 observations doing the work, and the standard errors should be read accordingly.
- **Power:** RD is data-hungry (previous frame); plan with **rdpower** (Cattaneo, Titiunik, and Vazquez-Bare 2019).

Covariates, Power, and What to Report (cont.)

- **Discrete running variables and heaping:** rounded ages, coarse test-score grids, or income bunched at round numbers do not invalidate the design, but they do invalidate the continuous- X asymptotics behind bandwidth selection and bias correction. Use a discrete-score method and say so.

Validity

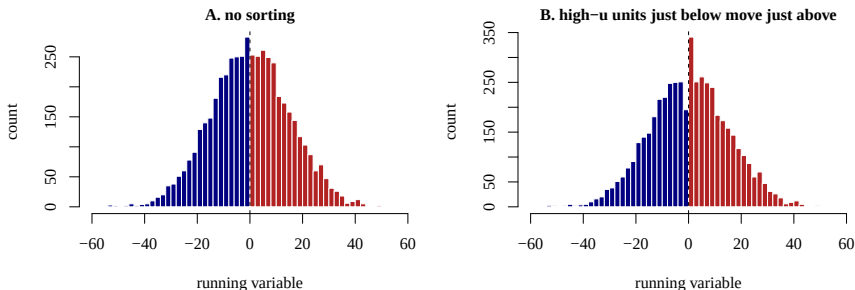
The Threat: Manipulation ★

Continuity fails if units can *precisely* control which side of the cutoff they land on, retaking a test until you pass, reporting income just under a threshold, a firm holding hiring just below a regulatory headcount.

- **Imprecise control is fine.** Lee (2008)'s insight is that if agents cannot control X to the last decimal, assignment near the cutoff is as good as random even though agents influence X .
- **Density test** (McCrary 2008; Cattaneo, Jansson, and Ma 2020): is the density of X continuous at c ? A spike on the advantaged side is direct evidence of sorting.
- **Covariate smoothness:** run the RD with *pre-determined covariates* as outcomes. They should show no jump. This is the single most persuasive check, and it is the RD analogue of a balance table.
- **Donut hole:** drop observations immediately around c , where manipulation would concentrate, and see whether the estimate survives.

What Sorting Does

Simulation: true effect of crossing = 0; 88 of the units landing in $[-2, 0)$, those with high unobserved u , move to $[0, 2)$ (figs/w9_sims.R).



- RD estimate (truth 0): 0.04 without sorting, **2.89** with it.
- Density test p -value: 0.071 without sorting, 0.00001 with it.

The movers bring their high u with them: m_0 now jumps at c , and the density shows the hole and the spike that produced it.

The Density Test, and the Senate

The test. Estimate the density of X at c separately from each side and compare:

$$T = \frac{\hat{f}_+(c) - \hat{f}_-(c)}{\widehat{\text{se}}} \approx N(0, 1) \text{ under no sorting.}$$

McCrary (2008) bins X and runs local linear regressions on the bin counts; Cattaneo, Jansson, and Ma (2020) fit a local polynomial to the empirical distribution function, with no bins (`rddensity`).

Senate: $T = -0.76$, $p = 0.45$; 49 elections lost by less than 2 points, 47 won by less than 2 (`figs/w9_senate.R`).

- A pass is weak evidence: sorting in *both* directions leaves the density smooth while changing who sits on each side.
- A failure is strong evidence, unless the spike has a mechanical cause such as rounding (next frames).

Covariate Balance on the Senate Data

Each pre-determined covariate as the outcome, `rdrobust` defaults, robust intervals
(figs/w9_senate.R):

covariate	mean below	jump	robust p
Democratic vote share, $t - 6$	47.86	3.88	0.09
Democratic vote share, $t - 12$	47.29	1.72	0.27
Democratic presidential vote share	43.95	-1.14	0.63
open seat	0.34	-0.14	0.09
state population (millions)	3.95	-0.18	0.82
House terms of the candidate	1.19	-0.17	0.56
Senate terms of the candidate	3.30	-0.19	0.90
midterm year	0.51	0.08	0.21
Democratic president	0.47	-0.08	0.37

- Two of nine at $p < 0.10$; about one expected by chance. The lagged vote is the one to watch: it is what moved the covariate-adjusted estimate.

Close Elections Under Scrutiny

The challenge (Caughey and Sekhon 2011): in U.S. House races 1942–2008 decided by less than 0.5 points, the party already holding the seat won 59 of 85 (69%). Bare winners also outspent their opponents more often. Imbalance *grew* closer to the cutoff, the opposite of what Lee's argument predicts.

The reply (Eggers, Fowler, Hainmueller, Hall, and Snyder 2015): over 40,000 close races (other House periods, statewide, state-legislative, mayoral, and nine other countries), the incumbent-party advantage in bare wins **appears nowhere else**.

- Precise control is possible where recounts, lawyers, and certification are contested; such places need balance checks that go beyond the density.
- The lesson is not “close elections are fine” or “are not”: the design is checked application by application, and the checks have found failures.

Placebo Cutoffs and Donuts on the Senate Data

Placebo cutoffs (one side only):

cutoff	estimate	robust p
-20	6.62	0.16
-15	-6.89	0.23
-10	-23.22	0.005
-5	-0.18	0.72
5	3.32	0.67
10	2.92	0.11
15	-1.22	0.65
20	-0.25	0.89

Donut: drop $|X_i| < \delta$:

δ	dropped	estimate	robust 95%
0	0	7.41	[4.09, 10.92]
0.25	9	6.94	[3.32, 10.45]
0.5	23	7.09	[3.20, 10.83]
1	43	6.58	[2.19, 10.60]
2	96	4.70	[0.21, 8.96]

- One placebo of eight rejects at 1%: with eight tests that happens about 8% of the time by chance. Look at the plot near -10 before deciding.
- The donut shrinks the estimate as it removes the closest races, which carry the most weight; with $\delta = 2$ the fit extrapolates over a 4-point gap.

Heaping: a Cautionary Tale

The design (Almond, Doyle, Kowalski, and Williams 2010): newborns under 1,500 g are classified “very low birth weight” and get more care. One-year mortality just below the cutoff is lower by about 1 point (-0.0095 , se 0.0048 ; mean 5.5% just above).

The problem (Barreca, Guldi, Lindo, and Waddell 2011): weights **heap** at round numbers (1,500 g; whole ounces), and heaped births are disproportionately from poorer, less healthy pregnancies. The heap sits exactly at the cutoff, on the treated side.

one-year mortality effect (85 g bandwidth)	estimate (se)
original	-0.0095
donut: drop multiples of 100 g	-0.0054 (0.0031)
donut: drop multiples of 100 g and of an ounce	0.0000 (0.0050)

Numbers from the working-paper version (IZA DP 5106). The original authors reply that the estimates fall but remain large (Almond et al. 2011). Heaping fails the density test for a reason unrelated to sorting.

When Sorting Is Present: Bounds (Gerard, Rokkanen, and Rothe 2020)

Suppose sorting only pushes units *up* past c . The density then has an excess just above c , and the excess is the movers:

$$\pi = \frac{f_+(c) - f_-(c)}{f_+(c)} = 1 - \frac{f_-(c)}{f_+(c)} \quad \text{share of units just above } c \text{ who moved there}$$

- The non-movers just above c are comparable to the units just below. We do not know *which* units above are movers, only how many.
- **Worst cases:** drop the top π share of outcomes above c , or the bottom π share. The two trimmed means bound $E[Y_i(1) \mid X_i = c, \text{non-mover}]$, and subtracting the mean just below bounds the effect.
- The trimming logic of bounds for sample attrition (Lee 2009). The bounds widen with π ; a density test that fails by a little costs a little.

Other Checks and Common Errors

- **Placebo cutoffs:** estimate at fake thresholds away from c , using only data on one side so the real discontinuity is not picked up. Significant “effects” there mean your procedure finds jumps everywhere.
- **Bandwidth sensitivity:** plot $\hat{\tau}$ against h over a wide range with robust intervals. A result that exists only at one bandwidth is not a result.
- **Do not use a high-order global polynomial** (Gelman and Imbens 2019). The implied weights on distant observations are large, oscillating, and indefensible; estimates wander with the degree.
- **Do not control for post-treatment covariates**, and do not interact covariates with D unless you mean to estimate heterogeneity.
- **Do not report only the conventional interval** at an MSE-optimal bandwidth.

Local Rand.

A Second Framework (Cattaneo, Frandsen, and Titiunik 2015)

Goal: treat a small window around c as an experiment, and analyse it as one.

Assumption: there is a window $W_0 = [c - w, c + w]$ in which

1. which units fall above c is as if decided by a coin flip, and
 2. $Y_i(d)$ does not depend on X_i inside the window.
- Then the difference in means inside W_0 estimates the effect, and a **Fisher randomization test** gives exact p -values: reshuffle who is above c among the window's units, recompute, compare.
 - **Continuity** extrapolates two curves to one point; **local randomization** assumes a whole window is an experiment. Condition 2 is stronger, and it is why w must be small (Sekhon and Titiunik 2017).
 - Natural when X is discrete with few values near c (ranks, ages in years), where curve fitting has little to work with.

Choosing the Window, Then Testing

Window: in nested windows, test balance on pre-determined covariates; take the largest window in which the smallest p -value stays above 0.15

(`rdlocrand::rdwinselect`). Senate, nine covariates (`figs/w9_senate.R`):

window	smallest balance p	below	above
± 0.500	0.225	9	14
± 0.625	0.208	13	17
± 0.750	0.065	15	22
± 1.000	0.043	17	25

Test in ± 0.625 : mean vote at $t + 6$ is 42.74 below and 52.36 above, a difference of **9.62**. Of 5,000 reshufflings of the 17 “above” labels among the 30 races, 0.12% give a difference this large: $p = 0.0012$.

Larger than the continuity estimate (7.41), from 30 races instead of 683. Cattaneo, Frandsen, and Titiunik (2015) use ± 0.75 (9.69 here).

Fuzzy

When the Rule Is Not Followed ★

Often crossing the cutoff changes the *probability* of treatment rather than determining it: eligibility is not take-up.

$$\lim_{x \downarrow c} \Pr(D_i = 1 \mid X_i = x) - \lim_{x \uparrow c} \Pr(D_i = 1 \mid X_i = x) \in (0, 1).$$

The fuzzy RD estimator is a ratio of two sharp RD estimates:

$$\tau_{FRD} = \frac{\text{jump in } Y \text{ at } c}{\text{jump in } D \text{ at } c} = \frac{\lim_{x \downarrow c} E[Y \mid x] - \lim_{x \uparrow c} E[Y \mid x]}{\lim_{x \downarrow c} E[D \mid x] - \lim_{x \uparrow c} E[D \mid x]}.$$

- This is exactly the Wald estimator of Week 7, with $\mathbb{1}\{X \geq c\}$ as the instrument, estimated locally.
- The numerator alone is the **intention-to-treat** effect of crossing the cutoff, and it is worth reporting on its own, it needs no first-stage assumption.

What the Ratio Estimates

Let $D_i(1)$, $D_i(0)$ be unit i 's treatment just above and just below c (compliers: $D_i(1) = 1$, $D_i(0) = 0$). As in Week 7:

jump in $Y = E[Y_i(D_i(1)) - Y_i(D_i(0)) X_i = c]$	continuity, for each type
$= E[(D_i(1) - D_i(0))(Y_i(1) - Y_i(0)) X_i = c]$	zero unless $D_i(1) \neq D_i(0)$
$= \Pr(C c) \tau_C(c) - \Pr(\text{defier} c) \tau_{\text{def}}(c)$	split by type
$= \Pr(C c) \tau_C(c)$	monotonicity: no defiers
jump in $D = \Pr(C c)$	same steps with D

- The ratio is $\tau_C(c)$: the effect for **compliers at the cutoff** (Hahn, Todd, and van der Klaauw 2001).
- “Continuity for each type” is the new assumption: the share of each type and its mean outcomes do not jump at c .

Estimation: Local 2SLS

Run the sharp RD twice, same kernel and same h , and divide:

$$\begin{aligned} Y_i &= \alpha_Y + \beta_Y Z_i + \gamma_Y (X_i - c) + \eta_Y Z_i (X_i - c) + \epsilon_i \\ D_i &= \alpha_D + \beta_D Z_i + \gamma_D (X_i - c) + \eta_D Z_i (X_i - c) + v_i, \quad \hat{\tau} = \hat{\beta}_Y / \hat{\beta}_D, \end{aligned}$$

with $Z_i = \mathbb{1}\{X_i \geq c\}$. Numerically the same: 2SLS of Y on D_i and $D_i(X_i - c)$, instrumented by Z_i and $Z_i(X_i - c)$, controlling for $(X_i - c)$.

- One bandwidth for both regressions; `rdrobust(y, x, fuzzy = d)` chooses it for the ratio.
- The standard error is the IV standard error of Week 7, computed locally.
- Report $\hat{\beta}_D$, the first stage, in the main table.

Fuzzy RD Inherits Everything From IV ★

- **Exclusion:** crossing the cutoff must affect Y only through D . If the threshold also triggers something else, a different school, a different caseworker, a label, exclusion fails.
- **Monotonicity:** no one is treated *because* they fell below the cutoff.
- **The estimand is doubly local:** the effect for compliers at the cutoff. Not the ATE, not the ATT, and not the effect for everyone near the cutoff.
- **Weak first stage** is a real risk: if take-up jumps by only a few percentage points, you are dividing by a small noisy number and all the Week 7 pathologies return. Report the size of the jump in D .

Weak First Stage: Anderson–Rubin at the Cutoff

The Week 7 idea carries over. To test $H_0 : \tau = \tau_0$, build

$$\tilde{Y}_i = Y_i - \tau_0 D_i, \quad \text{jump in } \tilde{Y} = \text{jump in } Y - \tau_0 \cdot \text{jump in } D,$$

which is zero exactly when τ_0 is the ratio. Run a *sharp* RD on $\tilde{Y}_i$; keep every τ_0 it does not reject.

- No division, so a small jump in D does not distort the test; the set can be wide or unbounded, and then it should be.
- Feir, Lemieux, and Marmer (2016): the usual fuzzy-RD t -test over-rejects when the jump in take-up is small; invert a test with the null-restricted standard error instead.
- Noack and Rothe (2024): the same construction with the bias of each sharp RD bounded as in honest intervals; valid with weak first stages, discrete running variables, and donuts.

Vietnam

In the Literature: Dell and Querubín (2018)

The question. Does force from above win hearts and minds? The effect of U.S. bombing on insurgency, local government, and civic life in South Vietnam, and of the Army's firepower doctrine against the Marines' "hearts and minds".

The data. The Hamlet Evaluation System (HES), 1969–1973: 169 monthly and quarterly questions on security, politics, and the economy, answered by U.S. advisers for nearly every hamlet. Outcomes: indices built from HES answers, armed-forces records of operations and attacks, and public opinion surveys.

Two designs.

1. **Bombing:** a fuzzy RD at the rounding thresholds of an algorithm that scored hamlet security; air strikes were targeted on the rounded score.
2. **Doctrine:** a boundary RD at the line between the Marines' I Corps and the Army's II Corps.

Numbers in this section are from NBER Working Paper 22395 (2016); the QJE version is *QJE* 133(2): 701–764.

The Score Nobody Saw

- The HES questions feed 19 submodels. Each starts from a flat prior over five security classes, A (5) to E (1), and updates by Bayes' rule using a matrix of conditional probabilities.
- The **posterior mean**, a continuous number between 1 and 5, is **rounded**: 3.49 becomes C, 3.51 becomes B. Rounded submodel scores are combined into one letter grade per hamlet.
- Air strikes were planned weekly by quota, with the hamlet grade among the inputs. Just below a rounding threshold, a hamlet is graded one letter worse than a near-identical one just above.
- The continuous scores were never printed; the conditional probabilities were classified. The authors rebuilt them from archived tapes and reproduce every rounded score.

Running variable: distance from the continuous score to the nearest of four thresholds (A/B, B/C, C/D, D/E); **instrument:** being below it. Hamlets near each threshold: 35%, 46%, 16%, 3%.

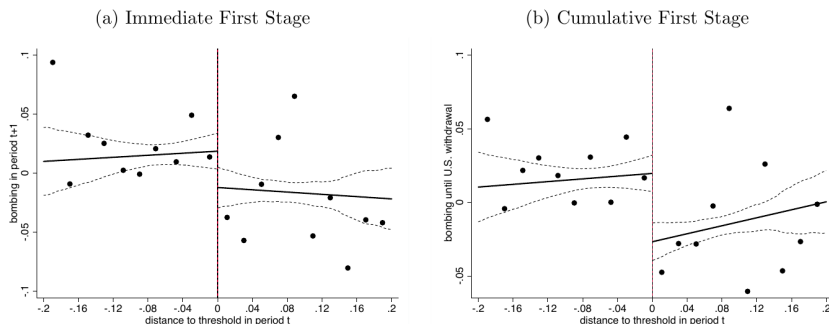
Dell and Querubín: the Estimating Equation

For hamlet h first near a threshold in quarter t (each hamlet once):

$$y_{h,t+n} = \gamma \text{below}_{ht} + \sum_{d=1}^4 \left(\delta_d D_{htd} + D_{htd} f_d(\text{dist}_{ht}) + D_{htd} f_d(\text{dist}_{ht}) \text{below}_{ht} \right) \\ + \alpha_t + X'_{ht} \beta + \epsilon_{ht}$$

- $D_{htd} = 1$ if threshold d is the nearest: a separate intercept and separate local linear slopes on each side of each threshold. This is the multi-cutoff pooling of the next section.
- α_t : quarter fixed effects; X_{ht} : indicators for every answer that entered the score, for precision.
- Local linear, Imbens–Kalyanaraman bandwidth; standard errors clustered by village.
- First stage: y = bombing; second stage instruments bombing with below_{ht} . The estimate is a LATE for hamlets bombed *because* they were rounded down.

Dell and Querubín: the First Stage



Their Figure 3(a)–(b). Horizontal: distance to the threshold; below the threshold on the left. Vertical: bombing, residualized.

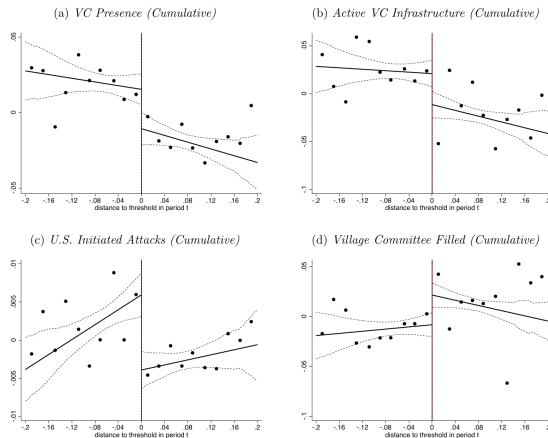
- Being just below raises the share of months with air or artillery strikes near the hamlet next quarter by 0.054 (0.014), from a mean of 0.28; $F = 14.9$. Through U.S. withdrawal: 0.044 (0.013), mean 0.26, $F = 11.5$.

Dell and Querubín: Checking the Design

No sorting	Scores were computed centrally and never printed; field staff did not know the probabilities. McCrary test: no jump in the density.
Continuity	Pre-period outcomes balanced (a few of many differ, e.g. South Vietnamese deaths). Bombing in the current and previous quarter does not jump: 0.011 (0.011), -0.001 (0.018).
Placebo year	In 1969 the score was computed but not used: no first stage, -0.016 (0.019).
Exclusion	No jump in other allocations (ground troops, U.S. operations, naval attacks, local forces): friendly forces nearby 0.016 (0.018).

- The 1969 placebo is the strongest check: same algorithm, same thresholds, no use of the score, no jump. Whatever moves bombing in 1970 is the score's use, not hamlet differences at the threshold.
- Exclusion is argued by measuring the other channels a lower grade could trigger.

Dell and Querubín: Reduced Forms



Their Figure 5(a)–(d), cumulative outcomes through withdrawal. Left of zero: rounded down, bombed more.

Dell and Querubín: Main Results

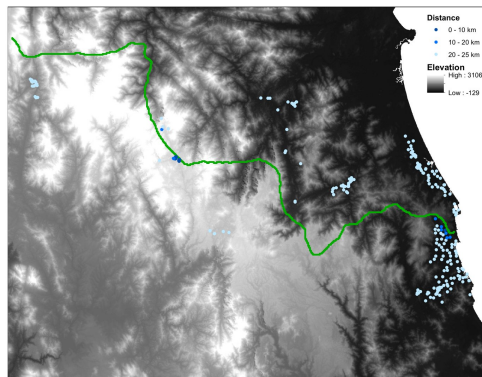
IV estimates, outcomes cumulative through U.S. withdrawal (their Tables 4, 6, 7):

outcome	IV (se)	mean	at mean bombing
security index	-0.642 (0.246)	0.68	-17 pts
armed Viet Cong present	0.571 (0.222)	0.19	+15 pts
Viet Cong infrastructure active	0.978 (0.384)	0.25	+25 pts
village committee filled	-0.798 (0.380)	0.84	-21 pts
local government collects taxes	-0.944 (0.443)	0.70	-25 pts
civic society index	-0.523 (0.248)	0.69	-14 pts

- **Reading:** a coefficient is the effect of going from no bombing to bombing in every month; the last column multiplies by mean bombing, 0.26 (e.g. $0.571 \times 0.26 = 0.15$).
- Bombing strengthened the insurgency it aimed at and weakened local government and non-communist civic life.

Dell and Querubín: a Boundary Design

Figure 7: Corps Region Boundary



Notes: This map plots hamlets near the Corps I-II boundary. See the legend for more details.

Their Figure 7: the I/II Corps boundary; hamlets within 25 km.

Marines (I Corps): small units in villages, “hearts and minds”; Army (II Corps): firepower.

$$y_{hs} = \alpha_0 + \alpha_1 \text{USMC}_{hs} + f(\text{lat}, \text{lon}) + G'_{hs}\beta + \alpha_s$$

- f : local linear in location; G : geography; α_s : boundary segment.
- 302 hamlets within 25 km, 64 village clusters.
- Results (Table 10): armed VC present -0.560 (0.075); VC attacks on hamlets -0.482 (0.061); education 0.243 (0.096); health 0.562 (0.198); security index 0.095 (0.110).

Dell and Querubín: What the Boundary Cannot Separate

- **Compound treatment.** More than doctrine changes at the line: the service, its soldiers, officer rotation (six months in combat for Army officers, up to a year for Marines). The authors grant these channels cannot be ruled out.
- **Balance is imperfect:** colonial military posts 0.300 (0.160); total roads 2.253 (0.816), more on the Marine side.
- **Few clusters:** 64 villages; attitudes come from 13 surveyed hamlets near the line, so that table is OLS within 100 km, not RD.
- **Comparison of the two designs:** the bombing RD has a single, measured treatment and a placebo year; the boundary RD compares two bundles of everything an army does.

The headline survives both: the firepower approach produced worse outcomes, on security and on the institutions it meant to build.

Extensions

Several Cutoffs: What Pooling Estimates

Group j faces cutoff c_j (with probability p_j). Pool by centring, $\tilde{X}_i = X_i - c_{J_i}$, and run one RD at $\tilde{X} = 0$:

$$\text{jump at } \tilde{X} = 0 = \sum_j \Pr(J_i = j \mid \tilde{X}_i = 0) \tau_j \quad \text{group jumps, mixed}$$

$$\Pr(J_i = j \mid \tilde{X}_i = 0) = \frac{f_j(c_j) p_j}{\sum_k f_k(c_k) p_k} \quad \text{Bayes' rule; } f_j: \text{ density in group } j$$

- The pooled effect is a weighted average of cutoff-specific effects, weights $\propto$ how many units sit at each cutoff (Cattaneo, Keele, Titiunik, and Vazquez-Bare 2016).
- In Dell and Querubín, 46% of hamlets sit near the B/C threshold and 3% near D/E: the pooled estimate is mostly about mid-grade hamlets.
- Report the cutoff-specific estimates when you can; they show whether one τ is a fair summary.

Away From the Cutoff

Goal: $\tau(x)$ for $x \neq c$, which continuity alone cannot deliver.

With two cutoffs $c_1 < c_2$, at $x \in (c_1, c_2)$ group 1 is treated and group 2 is not.

Assume the two control curves are parallel: $m_{0,1}(x) - m_{0,2}(x) = \Delta$ for all x .

$$\Delta = m_{0,1}(c_1) - m_{0,2}(c_1) \quad \text{both observed untreated at } c_1 \text{ (group 1 from the left)}$$

$$\tau_1(x) = m_{1,1}(x) - (m_{0,2}(x) + \Delta) \quad \text{the unseen } m_{0,1}(x) \text{ replaced}$$

(Cattaneo, Keele, Titiunik, and Vazquez-Bare 2021); the assumption is Week 10's parallel trends, with x for time.

Other routes:

- **Conditional independence:** given other predictors W , $Y(d)$ does not depend on X ; then W , not X , does the work away from c , and the assumption is testable on each side (Angrist and Rokkanen 2015).
- **The slope of the effect at c ,** $\tau'(c)$, is identified from the change in slopes; it answers “what if the threshold moved a little?” (Dong and Lewbel 2015).

Boundaries and Two Scores

Geographic RD: the running variable is location (lat, lon); the cutoff is a line (Keele and Titiunik 2015).

- The estimand is $\tau(b)$ at each boundary point b ; one number pools them, weighted by where units live along the line.
- **Distance to the boundary** as a single running variable mixes points that are close to the line but far from each other; boundary segment fixed effects, as in Dell and Querubín, compare neighbours.
- Recent work estimates $\tau(b)$ point by point, with location-based or distance-based local polynomials and valid intervals (Cattaneo, Titiunik, and Yu 2025; rd2d).
- **Compound treatment is the rule:** borders separate taxes, schools, police, and history at once. Name the bundle.

Two scores (pass both a reading and a math test): the treated region is a corner; each edge of the boundary has its own effect (Papay, Willett, and Murnane 2011; Reardon and Robinson 2012).

Regression Kink (Card, Lee, Pei, and Weber 2015)

A benefit $B_i = b(X_i)$ follows a known schedule that is continuous but *changes slope* at c (a benefit cap, a tax bracket). With $Y_i = y(B_i, X_i, U_i)$:

$$\frac{d}{dx} E[Y_i | X_i = x] = E \left[\frac{\partial y}{\partial B} \middle| x \right] b'(x) + \text{smooth terms} \quad \text{chain rule}$$

$$\underbrace{\lim_{x \downarrow c} \frac{d}{dx} E[Y | x] - \lim_{x \uparrow c} \frac{d}{dx} E[Y | x]}_{\text{kink in } Y} = E \left[\frac{\partial y}{\partial B} \middle| c \right] \underbrace{(b'(c^+) - b'(c^-))}_{\text{kink in } B} \quad \text{subtract}$$

- Effect = kink in Y / kink in B : the RD ratio with slopes for levels.
- Slopes are estimated far less precisely than levels; local quadratic fits and robust intervals are standard.
- Austrian unemployment benefits: higher benefits lengthen joblessness.

RD in Time

The running variable is the date; the treatment starts at t^* for everyone (Hausman and Rapson 2018).

- **No one sorts into a date**, but other things happen on dates: holidays, seasons, a second event. Check for coincident events.
- **Observations are not independent**: one time series, autocorrelated; inference must account for it.
- **Windows tend to be wide**: daily data with few days near t^* invite long windows, and then the RD is an interrupted time series with a polynomial trend.
- **Effects evolve**: anticipation before t^* , fading or building after. The jump at t^* is the short-run effect only.
- **Placebo dates** and a comparison group (Week 10) are the main defences.

Repeated Cutoffs: Dynamic RD

Goal: the effect of a treatment k years later when units that barely miss can try again (a district loses a bond vote, proposes again, passes).

The RD of “barely won at t ” on Y_{t+k} is an intention-to-treat effect, ITT_k . Let π_j be the RD jump in the chance of a bond at $t + j$ ($\pi_0 = 1$) and θ_k the effect of a bond k years after it passes:

$$ITT_k = \sum_{j=0}^k \pi_j \theta_{k-j} \quad \text{every bond induced at } t + j \text{ acts for } k - j \text{ years}$$

$$\theta_k = ITT_k - \sum_{j=1}^k \pi_j \theta_{k-j} \quad \text{solve from } \theta_0 = ITT_0 \text{ upward}$$

Hypothetical numbers: $ITT_0 = 10$, $ITT_1 = 12$, $\pi_1 = -0.4$ (losers pass a year later):
 $\theta_1 = 12 - (-0.4)(10) = 16$. The ITT understates the bond’s effect.

Dynamic RD: Applications and Assumptions

- California school bonds and house prices (Cellini, Ferreira, and Rothstein 2010); Wisconsin operating vs. capital referenda (Baron 2022).
- Assumes θ_k does not depend on a unit's history; Hsu and Shen (2024) identify the effects under weaker assumptions.
- Cumulative effect: $\sum_k \theta_k$. The ITT path is the effect of a close win, retries included; Bol et al.'s €180,000 below is of that kind.

Sociology

In the Literature: Bol, de Vaan, and van de Rijt (2018)

The question. Merton's Matthew effect: does early success cause later success, or do repeated winners just have more ability? Cumulative advantage is a cornerstone claim of stratification research and hard to separate from selection.

The data. The Dutch science foundation's early-career "Veni" grant (€250,000): review scores and decisions for recent PhDs, followed for up to 16 years.

The design. A hard threshold: every winner scored above every non-winner. Running variable: the applicant's **rank** relative to the funding line in each competition; compare the two ranks just above (+1, +2) with the two just below (-1, -2), with exact tests. A sharp RD with a discrete running variable, analysed as a local experiment.

Outcomes. Later grants (the mid-career "Vidi" and three others), applications for them, publications and citations, a full professorship.

Bol, de Vaan, and van de Rijt: Checking the Design

No sorting	Applicants cannot place their score; ranks are set by panels. A density test is uninformative: ranks are evenly spaced by construction.
Continuity	Publications, citations, and H-index <i>before</i> the competition rise smoothly with rank, with no jump at the line.
Placebo cutoffs	No difference in later outcomes between any other pair of neighbouring ranks.
Specification	A parametric RD (supplement) gives the same size and significance.

- Only rank is used, not the score; the comparison of $\pm 1, 2$ is the local-randomization window, chosen before looking.
- Productivity *after* the grant also shows no jump: the grant did not buy measurably better science, which rules out one mechanism.

Bol, de Vaan, and van de Rijt: Main Results

Ranks +1/ + 2 against -1/ - 2:

outcome	just below	just above	test
won a mid-career grant	10%	26%	$P < 0.001$
applied for a mid-career grant	40%	59%	$P < 0.001$
later funding, 4 years on		+€40,000	$P < 0.01$
later funding, 8 years on		+€180,000	$P < 0.01$
full professor in 2018	19%	28%	$P < 0.05$

- The near-winners got more than twice the later money of near-losers, excluding the Veni itself; this is 40% of the gap between best and worst.
- **Participation:** near-losers stop applying. A critique reran the RD on Vidi applicants only; that conditions on applying, a post-treatment variable (the Week 5 collider). Winning raised applications by about half and wins by 100–150% (Bol et al. 2019 reply).

In the Literature: Legewie (2016)

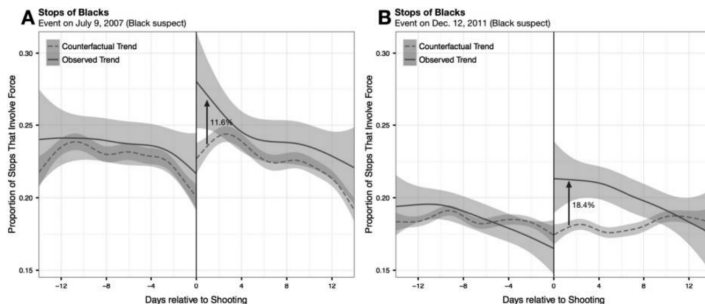
The question. Do salient events trigger periods of more racially biased policing? Two NYPD officers were shot by Black suspects (July 9, 2007; December 12, 2011).

The data. About 3.9 million NYPD pedestrian stops, 2006–2012. Outcome: any physical force in the stop (16% of stops of whites, 22% of Blacks, 24% of Hispanics).

The design. Time is the running variable; the shooting is the cutoff. The main analysis compares stops in the three days after each shooting with matched stops (same precinct, race, place, time of day, suspect characteristics) from the year before; an RD in time with local linear fits is the robustness check.

three days after, log-odds of force (se)	White	Black	Hispanic
December 2011	−0.30 (0.16)	0.20 (0.05)	−0.10 (0.08)
July 2007	−0.01 (0.14)	0.16 (0.06)	0.02 (0.08)

Legewie: Results and Checks



His Figure 3: force in stops of Black suspects, observed against counterfactual trend.

- Force against Blacks up 16% (21.9% to 25.4%) in 2011, 13% in 2007; no change for whites or Hispanics; it fades in about 3.5 and 10 days.
- **Placebo events:** shootings by white or Hispanic suspects show no effect. Among 800 random sets of dates, 1.3% match the Black-suspect pattern.
- **RD in time:** estimates across bandwidths stay positive; very short windows shrink the 2007 effect, as news needs time to spread.

Practice

Reading and Reporting an RD ★

1. **The picture first.** Binned means of Y against X , with the fitted local polynomials and the cutoff. If the jump is not visible, be sceptical of the number.
2. The bandwidth, how it was chosen, and the effective n on each side.
3. Robust bias-corrected confidence intervals.
4. The density test and the covariate-smoothness checks.
5. A bandwidth-sensitivity plot and placebo cutoffs.
6. For fuzzy designs, the size of the jump in treatment take-up, and the intention-to-treat effect alongside.
7. An explicit statement that the estimand is local to the cutoff.

Looking Ahead

- RD buys extraordinary credibility at the cost of a very local estimand. The identifying variation is visible in a scatterplot, and the main threats (sorting, bandwidth dependence) are checkable.
- Note the family resemblance: sharp RD is a local experiment; fuzzy RD is local IV. The tools recombine rather than multiply.
- **Next week:** designs that use variation over *time* rather than across a threshold.
- Note again how the estimand was chosen by the design rather than by us. Saying so plainly is part of reporting an RD honestly.

Problem sets

Problem Set 3 is due Monday October 19. Problem Set 4 will be assigned next Tuesday (October 27) and is due Monday November 9; it covers Weeks 7 and 9.

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